Experimental Design and Implementation Analysis

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Pre-Analysis Plan

Can Directing Agricultural Extension Services and Microcredit Exclusively to Women Shift Their Perceived Role in Nigerian Farm Households?

1. Introduction

The gender gap in agricultural productivity represents one of the most persistent and economically consequential forms of inequality in the developing world. In Nigeria, its dimensions are particularly acute. Drawing on the 2018–2019 Nigeria General Household Survey, the World Bank estimates that female plot managers produce roughly 30 percent less per hectare than male farmers and, in southern Nigeria specifically, generate 25 percent less output per hectare after conditioning on observable household characteristics (World Bank, 2023). The aggregate cost of this difference is estimated at $2.3 billion annually, or about 0.6 percent of Nigeria's GDP, with broader spillover gains of up to 2 percent of GDP if the productivity gap were closed (World Bank, 2023).

A cross-country FAO analysis similarly finds that the gender gap in agricultural labor productivity in Nigeria is large relative to other sub-Saharan African contexts (Palacios-López & López, 2015). What makes this gap particularly striking is that it cannot be explained fully by observable differences in endowments. Mukasa and Salami's Oaxaca–Blinder decomposition for smallholder rice farmers in southwest Nigeria suggests that nearly 77 percent of the productivity difference between male and female farmers is attributable to unexplained structural factors rather than to differences in endowments (Mukasa & Salami, 2015). The implication is theoretically significant: interventions that target only the observable aspects of the gap, such as inputs or credit, are unlikely to generate durable gains if the underlying institutional and social mechanisms remain unchanged.

This paper instead argues that a central structural mechanism is the institutional signal of economic recognition: productive resources such as land, credit, inputs, and knowledge are more often directed toward men than toward women, reinforcing the perception that men are the farm's legitimate economic agents while women are dependents rather than contributors (Oseni et al., 2015). This argument builds on a large intra-household literature. Duflo and Udry (2004) show in Côte d'Ivoire that spouses farm separate plots and do not pool resources optimally, while Goldstein and Udry (2008) show in Ghana that insecure land rights suppress investment and disproportionately constrain women. Taken together, this literature suggests that the identity of

the resource recipient shapes productive outcomes in ways that aggregate quantities of inputs alone cannot capture.

The literature also shows, however, that access to resources does not automatically translate into bargaining power. Swaans et al. (2013) note that women's increased participation in agricultural value chains does not necessarily improve their control over the resulting income, and male household members may capture commercially successful female-managed activities. A primary reason many interventions fail is their exclusive focus on visible outputs rather than the structural drivers that sustain them (Farnworth & Colverson, 2015). The relevant intervention, therefore, is not simply providing resources to women but shifting the intra-household perception of women toward that of productive labor and recognized economic contribution.

The present study is designed to test that mechanism directly. It asks whether directing agricultural extension services and microcredit exclusively to women increases women's decision-making authority and control over farm resources, and whether these shifts alter their perceived role as productive contributors within the household. Women's economic participation and plot-level productivity are treated as important secondary outcomes. The experimental sites span rural northern and southern Nigeria, allowing the design to exploit meaningful variation in gender norm intensity across the Purdah/non-Purdah regional gradient.

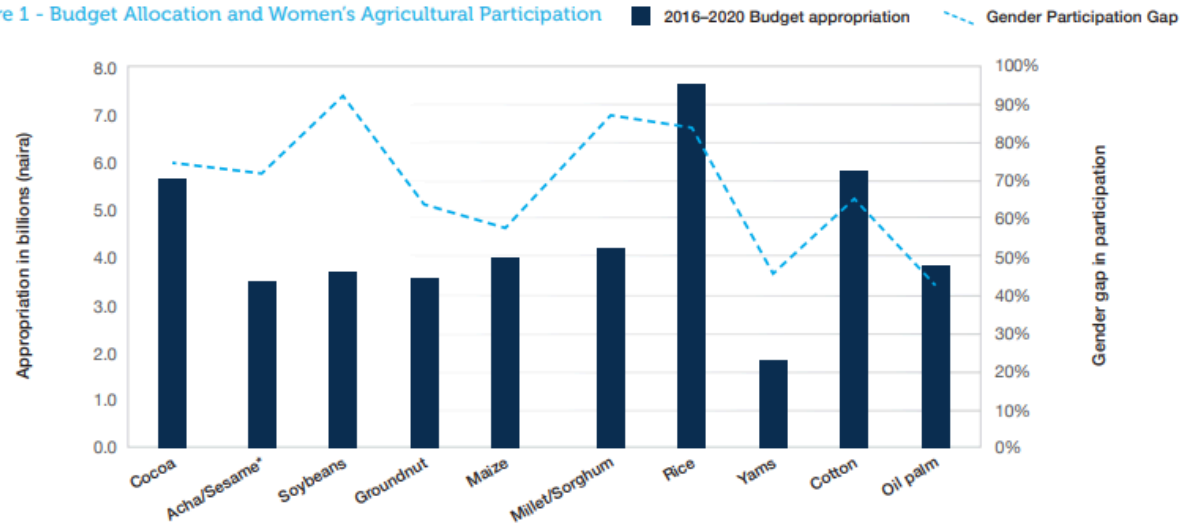
1.1 Background

Agriculture accounts for roughly 24 percent of Nigeria's GDP, and smallholder farmers make up approximately 80 percent of the agricultural population while producing about 90 percent of national output. Agricultural activity is heavily concentrated in the rural north, where more than 80 percent of households are engaged in farming, compared with roughly 50 percent in the urban south (Bello et al., 2021). Women comprise 36.2 percent of the agricultural labor force according to the 2023 Nigerian Labour Force Survey, yet they are 10 percent less likely than men to be active in the agricultural sector and 25 percent less likely to serve as primary plot managers (World Bank & International Labour Organization, 2023).

The productivity gap between male and female plot managers is both pronounced and multidimensional. Oseni et al. (2015) document that female-managed plots in northern Nigeria are, on average, 42 percent smaller than male-managed plots, that women rely more heavily on female household labor per hectare, and that the gender productivity gap is wider in the north than in the south. The World Bank identifies three co-constitutive drivers of this gap: lower use of modern inputs, concentration in lower-value crop production, and weaker access to hired labor and mechanization (World Bank, 2023). Women are, for example, 38 percent more likely to cultivate roots and tubers and 19 percent less likely to cultivate higher-value cereals (Fashogbon et al., 2023).

Drawing on an analysis of 2016–2020 Federal Ministry of Agriculture and Rural Development appropriations by the Nigeria Gender Innovation Lab, we find that the value chains receiving the greatest public investment—such as cotton, rice, sorghum, and cocoa—are also among those with the largest gender participation gaps (Fashogbon et al., 2023). Across these value chains, physical inputs and extension contacts remain disproportionately allocated to male farmers, reinforcing existing asymmetries in knowledge, capital, and productive opportunity (World Bank, 2023).

Figure 1 - Budget Allocation and Women's Agricultural Participation



* = The indicated crop was included in the budget analysis, but not the earlier GHS analysis.

2. Literature Review

This review examines the theoretical and empirical foundations for understanding gendered productivity gaps in agriculture, with a focus on the intersection of intra-household allocation, institutional signaling, and social norms in the Nigerian context.

Theoretical Frameworks of the Household and Resource Allocation

The primary theoretical lens for this study is the shift from unitary to collective household models. Collective approaches reject the assumption that households behave as single decision-making units with fully pooled resources and instead emphasize distinct preferences, bargaining positions, and sharing rules among household members (Bourguignon & Chiappori, 1992). In rural sub-Saharan Africa, this framework is especially useful because men and women often manage separate plots and retain partial control over income streams. Within this framework, the identity of the resource recipient is not incidental; it is central to how resources are used, how labor is allocated, and how authority is recognized. This logic directly informs the study's primary outcomes—women's decision-making authority and control over farm revenue—because these are the dimensions collective models predict should shift when productive resources are deliberately routed to women.

Institutional Signaling and Information Asymmetry

Institutional signaling provides a behavioral account of why gendered resource gaps persist. When public institutions such as extension systems interact primarily with men, they communicate that men are the legitimate agricultural actors and that women occupy a subordinate role. This signal can suppress women's bargaining power within the household even when women contribute substantial labor. Agricultural extension in Nigeria has historically been male-biased, reflecting a broader tendency for women to be underrepresented in formal market nodes and managerial roles (World Bank, 2023).

Recent experimental evidence suggests that these informational asymmetries are not fixed. In Uganda, providing agricultural information directly to female co-heads improved women's agricultural knowledge, participation in farm decision-making, adoption of recommended practices and inputs, and yields on the plots they managed (Lecoutere et al., 2023). In Mozambique, training female instructors reduced gender bias in the diffusion of agricultural techniques to women farmers (Kondylis et al., 2016). These findings suggest that the delivery channel itself matters. By designating women as the visible recipients of extension knowledge, institutions may begin to shift women's perceived productive role within the household.

The Synergy of Capital and Knowledge

Information is a vital input, but it is rarely sufficient on its own. The development literature shows that male household members may capture the benefits of female-led activities once those activities become commercially meaningful. For this reason, integrated interventions that combine capacity building with financial capital are increasingly favored. In Nigeria, the Nigeria for Women Project provides a useful implementation analogue: it combines women's group formation, confidence- and skills-building, and phased livelihood support rather than treating training and finance as isolated, one-off inputs (World Bank, 2018, 2024).

The rationale for pairing extension with credit is also supported by evidence from Ethiopia. Buehren et al. (2019) show that gender-inclusive extension can improve women's access to services, increase economic participation, and encourage more market-oriented production, but it does not automatically eliminate preexisting gender gaps in agricultural outcomes. In this study, microcredit directed explicitly to women through visible, designated channels is therefore conceptualized as an additional and reinforcing signal of economic recognition rather than as a stand-alone anti-poverty tool.

Social Norms and Regional Variation

The effectiveness of these interventions is likely to depend on the intensity of local social norms. In Nigeria, the Purdah/non-Purdah gradient creates meaningful variation in gender norm rigidity, with northern Nigeria more likely to be characterized by female seclusion and mobility restrictions. These constraints shape not only women's public participation but also the types of

crops they cultivate and the terms under which they access labor, markets, and information (Oseni et al., 2015; World Bank, 2023).

To evaluate interventions in such contexts, researchers increasingly rely on measures that move beyond simple production outcomes. The project-level Women's Empowerment in Agriculture Index (pro-WEAI) captures decision-making authority, control over revenue, and relational agency in agricultural settings (Malapit et al., 2019). It thereby provides a more comprehensive basis for assessing whether institutional interventions can disrupt entrenched gender hierarchies in rural Nigeria.

3. Experimental Design

3.1 Research Design

This study proposes a village-level randomized controlled trial to test whether directing productive resources exclusively to women changes their economic participation and perceived role within farm households in Nigeria. The intervention has three arms. Villages in the control group will continue under the status quo and will not receive study-provided gender-targeted services. Villages in Treatment 1 will receive women-only agricultural extension services. Villages in Treatment 2 will receive the same women-only agricultural extension services together with women-only microcredit eligibility. By comparing Treatment 1 to the control group, the study identifies the effect of directing agricultural knowledge to women. By comparing Treatment 2 to Treatment 1, the study isolates the additional effect of coupling knowledge with female-targeted financial capital. Comparing Treatment 2 to the control group captures the joint effect of both interventions.

This design is motivated by the persistence of gender gaps in Nigerian agriculture, especially women's lower access to extension services and other productive resources (World Bank, 2023). It also reflects the logic of women-focused livelihoods programs in Nigeria that combine group formation, norm-sensitive implementation, capacity building, and phased livelihood support, rather than delivering training and financial support as isolated, one-off inputs (World Bank, 2018, 2024).

The unit of randomization will be the village rather than the individual or household. This choice is appropriate for both conceptual and practical reasons. First, the mechanism under study is not purely individual. The intervention is designed to shift household perceptions of women's productive role, and these perceptions are likely to be shaped not only by private resource allocation but also by community-level visibility, participation in women's groups, and local norms regarding women's engagement in agriculture. Second, randomizing within villages would create substantial contamination risks, as information from extension sessions and knowledge about women's credit eligibility could diffuse across households. Village-level

assignment therefore offers the cleanest design for minimizing within-village spillovers while preserving the social visibility of the intervention.

The sample will consist of rural farming households in selected northern and southern Nigerian states with at least one adult woman engaged in agricultural production. Restricting the sample to such households ensures that the intervention is targeted toward women with a plausible opportunity to use agricultural information and financial resources. The inclusion of both northern and southern sites is central to the design because the study seeks to assess how treatment effects vary across settings with different levels of gender norm rigidity.

3.2 Sampling and Randomization

The study will first construct a sampling frame of eligible rural villages in the selected study states. Within each village, a baseline census or screening exercise will identify households containing at least one adult woman engaged in farming. Villages will then be randomly assigned to one of the three study arms. To improve balance and increase precision, randomization will be stratified by state or region and, if baseline data permit, by a village-level proxy for gender norm rigidity. Possible proxies include reported restrictions on women's mobility, the prevalence of female seclusion practices, or baseline attitudes regarding women's participation in farm decision-making.

Following randomization, the research team will conduct a baseline household survey prior to intervention rollout. The survey will gather information on household demographics, agricultural production, input use, credit access, women's labor participation, decision-making authority, control over farm income, and attitudinal measures related to women's legitimacy as economic actors within the household. The same households will be followed over time to form a panel dataset. Given the study's interest in both short-run behavioral changes and slower-moving changes in intra-household perceptions, data collection will occur at baseline, an early midline, a post-harvest midline, and an endline. A reasonable schedule is baseline before rollout, a first midline after approximately six months, a second follow-up after one full agricultural production cycle, and a final endline after roughly twenty-four to thirty months.

The final number of villages and households will be determined by formal power calculations. Because treatment is assigned at the village level, minimum detectable effects will depend on the intra-cluster correlation of the primary outcomes as well as expected attrition over the study period. The present draft therefore treats the sample counts that follow as preliminary rather than fixed.

3.3 Intervention

In Treatment 1 villages, eligible women will be invited to participate in women-only agricultural extension groups. These groups will receive regular training sessions focused on practical agricultural production and farm management. The curriculum will include modules on

crop planning, input use, basic recordkeeping, post-harvest handling, and market-oriented decision-making. To maximize women's participation, sessions will be delivered in accessible village locations and scheduled at times that reduce conflicts with domestic labor burdens, consistent with prior work showing that participation depends in part on reducing barriers tied to meeting design, timing, and location (Johnson et al., 2018). Where feasible, training will be delivered or co-delivered by female facilitators or female extension agents. This feature is supported by evidence that women-focused project strategies often rely on women-only groups and women staff, and by experimental evidence from Mozambique showing that female instructors can reduce gender bias in the diffusion of agricultural techniques to women farmers (Johnson et al., 2018; Kondylis et al., 2016).

The intervention is intentionally designed so that the woman, rather than another household member, is publicly recognized as the direct recipient of productive knowledge. This direct-targeting logic is important because evidence from Uganda suggests that providing agricultural information directly to women, rather than relying on male household members as intermediaries, can improve women's agricultural knowledge, decision-making participation, adoption of recommended practices and inputs, and yields on the plots they manage (Lecoutere et al., 2023).

Treatment 2 villages will receive the same women-only extension program and, in addition, eligible women will become the sole household members eligible for study-facilitated microcredit. To preserve the sequencing of the intervention and improve the likelihood that credit is used productively, credit eligibility will begin only after an initial extension period. This phased design mirrors women-focused livelihoods programs that first build group capacity, financial literacy, and implementation readiness before expanding access to livelihood finance (World Bank, 2018, 2024). The credit component will be administered through a partner microfinance institution and linked to the woman as the named recipient. Where operationally feasible, disbursement will be made into a bank account, mobile money account, or other designated financial mechanism explicitly associated with the woman recipient.

This design is intended to make both knowledge and capital visibly female-directed, thereby strengthening the intervention's hypothesized effect on women's economic agency and household recognition. The distinction between Treatment 1 and Treatment 2 is also theoretically important. Evidence from Ethiopia suggests that gender-inclusive extension systems can improve economic participation and market-oriented production, but extension alone may not be sufficient to eliminate preexisting gender gaps in agricultural outcomes, which supports the value of testing whether an added credit component generates stronger effects than extension alone (Buehren et al., 2019).

The control group will receive no study-provided extension or credit. Households in control villages will continue to access whatever services are normally available in their localities. This

preserves a realistic counterfactual and allows the study to estimate the effect of adding gender-targeted resources relative to prevailing institutional conditions.

3.4 Outcomes and Measurement

The study will focus on two primary outcomes. The first is a women's agricultural decision-making index, measuring the extent to which women report meaningful input into decisions regarding crop choice, input purchase, labor allocation, production practices, and marketing. The second is women's control over farm revenue, capturing whether women can decide how agricultural earnings are used, saved, or reinvested. These outcomes align directly with the study's theory of change and with collective household models, which predict that directing productive resources to women should alter intrahousehold authority and control even when aggregate household resources do not change immediately (Bourguignon & Chiappori, 1992).

Secondary outcomes will include women's agricultural labor participation, plot-level productivity, input adoption, and market participation. The study will also measure a perceived productive role index, constructed from attitudinal and relational survey items. These may include whether the woman is identified by herself and by her spouse as a principal contributor to household production, whether her agricultural knowledge is viewed as credible, and whether she is seen as entitled to participate in decisions over agricultural resources and earnings.

To improve conceptual consistency, the household survey will adapt items from established women's empowerment instruments used in agricultural project settings, particularly the project-level Women's Empowerment in Agriculture Index (pro-WEAI), while tailoring the wording to the Nigerian context and the study's mechanism of interest (Malapit et al., 2019). Using a survey-based empowerment framework is especially important here because the study is not only interested in production outcomes, but also in the distribution of decision-making authority and control over income within the household (Malapit et al., 2019).

3.5 Empirical Strategy

The main analysis will estimate intent-to-treat effects using village-level treatment assignment. A parsimonious empirical specification is:

$$Y_{iv1} = \alpha + \beta_1 Treat_{1v} + \beta_2 Treat_{2v} + \gamma Y_{iv0} + \delta_s + \varepsilon_{iv}$$

where Y_{iv1} is the follow-up outcome for household i in village v, T_{1v} indicates assignment to the women-only extension arm, $T2_v$ indicates assignment to the women-only extension plus microcredit arm, Y_{iv0} is the baseline value of the outcome, and δ_s denotes randomization-stratum fixed effects. Standard errors will be clustered at the village level.

This specification yields three key comparisons: β_1 estimates the effect of women-only extension relative to the control group; β_2 estimates the joint effect of women-only extension and microcredit relative to the control group; and a test of $\beta_2 - \beta_1$ identifies the marginal contribution of microcredit beyond extension alone. Where multiple follow-up rounds are pooled, the analysis may be extended to include survey-wave fixed effects and treatment-by-wave interactions to examine treatment dynamics over time.

3.6 Heterogeneity, Spillovers, Compliance, and Attrition

A central contribution of the study is its pre-specified heterogeneity analysis. Because the choice of Nigeria is partly motivated by regional variation in gender norm rigidity, the paper will test whether treatment effects differ across northern and southern sites and, where feasible, across baseline levels of local gender norm intensity. The direction of this heterogeneity is theoretically ambiguous. In more restrictive settings, stronger mobility constraints and tighter norms may reduce women's participation in public extension activities and limit women's effective control over credit. At the same time, because baseline levels of recognized female agency may be lower in such settings, a successful intervention could generate larger marginal changes. The paper therefore hypothesizes that treatment effects will differ by region, but does not impose a unidirectional prior.

Potential spillovers remain an important concern. Village-level randomization reduces within-village contamination by assigning entire communities to a single status. However, information and behavioral spillovers may still occur across neighboring villages through kinship networks, social ties, or local markets. To address this, the study will document village proximity and respondents' exposure to participants in other communities. As a robustness check, the analysis may exclude villages located very near treated communities or estimate whether control households with greater contact to treated villages display indirect treatment effects.

Compliance and treatment fidelity are also central to interpretation. Not all eligible women in Treatment 1 villages may attend extension sessions, and not all women in Treatment 2 villages may retain effective control over microcredit once it is disbursed. Because the study's theory hinges on women being recognized as the recipients of resources, low take-up or male appropriation of credit would weaken the institutional-signaling mechanism. The research team will therefore track attendance, session completion, receipt of credit, the identity of the account holder or named recipient, and self-reported control over loan use. The primary analysis will remain intent-to-treat, but these implementation measures will be used to assess treatment fidelity and to interpret attenuated estimates.

Finally, the study will take attrition seriously. Because mobility, migration, and seasonal labor movement are common in rural settings, panel attrition could bias estimated effects if it differs systematically across treatment arms. To mitigate this risk, the research team will collect

detailed contact information at baseline, including alternate contacts and GPS coordinates where feasible, and will implement structured tracking protocols at each follow-up. The analysis will test for differential attrition by treatment status and, if needed, report robustness checks using inverse probability weighting or bounding approaches.

3.7 Ethics and Informed Consent

Because the study involves village-level randomization and sensitive questions about gender, authority, and household resource control, the research team will seek approval from the relevant institutional review board before field implementation. All respondents will provide informed consent prior to participation. Survey protocols will be designed to protect privacy, minimize the risk of household conflict, and ensure that questions concerning control over credit or intra-household decision-making are asked in settings that preserve confidentiality.

The study will also account for context-specific ethical constraints, especially in northern sites where women's mobility and public participation may be more restricted. Enumerators will receive training on gender-sensitive interviewing, confidentiality, and referral procedures if respondents disclose distress or interpersonal conflict. Where local norms require additional permission for women's participation in public activities, the project will distinguish clearly between community entry protocols and individual consent so that household- or community-level gatekeeping does not substitute for a respondent's own voluntary agreement to participate.

4. Data and Power Calculation

4.1 Data Sources

The study will draw on two complementary data sources. First, the 2018–2019 Nigeria General Household Survey, conducted by the National Bureau of Statistics in collaboration with the World Bank, will serve as the basis for baseline parameter estimation and preliminary power calculations. The survey is nationally representative and provides plot-level data disaggregated by the gender of the plot manager, making it well suited for establishing baseline values of key outcomes in rural Nigerian farming households.

Second, the study will generate original primary data through a panel household survey administered at baseline, midline, and endline. The survey instrument will collect information on women's agricultural labor participation, plot-level productivity, input adoption, decision-making authority within the household, control over farm revenue, and attitudinal measures related to women's perceived productive role. Survey items measuring empowerment and decision-making will be adapted from the project-level Women's Empowerment in Agriculture Index (pro-WEAI), with wording tailored to the Nigerian context. The baseline survey will be conducted prior to intervention rollout and will also serve as the basis for verifying balance across treatment arms.

4.2 Hypotheses

The study tests three pre-specified hypotheses corresponding to the key comparisons in the experimental design.

H1: Women-only agricultural extension will increase women's decision-making authority and control over farm revenue relative to the control group. Women's economic participation and productivity are expected to improve as secondary outcomes. This hypothesis follows from evidence that directing agricultural knowledge specifically to women, rather than relying on male household members as intermediaries, can strengthen women's role in agricultural decision-making and adoption.

H2: Women-only agricultural extension combined with women-only microcredit eligibility will generate larger increases in women's decision-making authority, control over farm revenue, and perceived productive role than extension alone. This hypothesis rests on the argument that productive knowledge alone is insufficient to alter women's recognized role within the household without complementary access to financial capital.

H3: Treatment effects will differ by region, though the direction is theoretically ambiguous. In northern Nigeria, stricter gender norms may dampen effects by reducing women's participation in public sessions and limiting effective control over credit. Alternatively, if the intervention succeeds in a lower-baseline environment, it may generate larger marginal changes in recognized economic agency than in southern sites.

4.3 Power Calculations

We conduct power calculations to determine the minimum sample size required to detect treatment effects on our primary outcomes. Following standard practice, we assume a two-sided test with a significance level of 5% ($\alpha = 0.05$) and statistical power of 80% ($1 - \beta = 0.80$).

Because treatment is assigned at the village level, we must account for the intra-cluster correlation (ICC) of outcomes within villages. The required sample size per arm under cluster randomization is given by:

$$n = (1 + (m - 1) \times \rho) \times 2(z_{\frac{2}{\alpha}} + z_{\beta})^2 \times \frac{\sigma^2}{\delta^2}$$

where m is the average number of households per village, ρ is the ICC, σ is the standard deviation of the outcome, and δ is the minimum detectable effect (MDE).

For our primary outcome of women's agricultural labor participation, we draw on the 2018-19 Nigeria General Household Survey (GHS), which documents a baseline female agricultural labor participation rate of approximately 37% in rural Nigeria. For a binary outcome,

the standard deviation is given by $\sigma = \sqrt{p \times (1 - p)}$, where p is the baseline participation rate. We set $p = 0.37$, yielding $\sigma = \sqrt{(0.37 \times 0.63)} \approx 0.48$.

We set the MDE at 10 percentage points, consistent with effect sizes reported in comparable women-targeted agricultural interventions in Sub-Saharan Africa. Assuming an average of 20 households per village and an ICC of 0.10, which is a conservative assumption for rural Nigerian villages, the required number of villages per arm is approximately 15, corresponding to roughly 300 households per arm and a total sample of 900 households across the three study arms.

These calculations are preliminary. Final sample size determination will incorporate ICC estimates from the baseline survey and may be adjusted to account for expected attrition over the study period. Because 15 villages per arm is a relatively small number for a cluster-randomized design, the estimate should be treated as an initial planning benchmark rather than a fixed field target.

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